

Scalars, vectors, contact and non-contact forces

Forces
Paper 2 · Combined and Separate
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	Define a scalar quantity.	A quantity that has magnitude only.	☐
2	Give three examples of scalar quantities.	Any three from: energy, time, mass, distance, speed, temperature	☐
3 ★	Define a vector quantity.	A quantity that has both magnitude and direction.	☐
4	Give three examples of vector quantities.	Any three from: force, displacement, velocity, acceleration, momentum	☐
5	State how a vector is represented on a diagram.	By an arrow.	☐
6	Explain why an arrow is a suitable way to represent a vector.	Its length shows the magnitude and its direction shows the direction.	☐
7	State what a force is.	A push or pull on an object due to an interaction with another object.	☐
8	What is a contact force?	A force between objects that are physically touching.	☐
9 ★	Give three examples of contact forces.	Any three from: friction, air resistance, tension, normal contact force	☐
10	What is a non-contact force?	A force between objects that are physically separated.	☐
11	Give three examples of non-contact forces.	Gravitational force (weight), electrostatic force, magnetic force	☐
12	When two objects interact, state which of them experiences a force.	Both: the forces are equal in size and opposite in direction.	☐